

continuation

Continuation Document

Revision: 56 **Last modified:** 2026-06-01T17:28:49Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

§1: Recently Completed Work (last 10 commits)

Commit	Message
467a6dc	Foundation wave 10: Vault secret injection + rotation (KV v2, no-downtime refresh), WireGuard mesh segmentation policy engine (DENY-wins label selectors + enforcement ruleset), Backup service (etcd clientv3 snapshot + pg_dump/restore state-match) — integration-proven vs real vault/etcd/postgres. GATE: fixed 2 non-compiling integration files (unused ctx, unused require import)
f1e920b	Foundation wave 9: mark 4 items Completed (HXC-1104/1116/1127/1132) — integration-proven vs real redis/mTLS/opa/HTTP; CLAUDE-2 macOS CPU utilization now real
7634fbb	Foundation wave 9: Redis store (routing TTL + NodeEvent pub/sub, go-redis/v9), SPIFFE SVID issuance + mTLS identity (go-spiffe/v2), OPA policy engine + HelixConstitution scheduling (open-policy-agent/opa), metrics collector scrape endpoint — integration-proven vs real redis/mTLS/opa/HTTP. GATE FIXES: tmux ControlModeAttach EOF-before-drain race; pkg/resources DarwinReader now samples REAL macOS CPU utilization via top (CLAUDE-2: no more 0%% on macOS); redis test mutex-copy vet violation
51cea15	

Commit	Message
6c825df	Foundation wave 8: mark 5 items Completed (HXC-1108/1109/1118/1120/1121) — integration-proven vs real NATS/Kafka/HTTP Foundation wave 8: NATS JetStream stream-config (5 streams + EnsureStreams), Kafka topic-config (partitions/replication/retention + EnsureTopics), HelixQA challenge runner + DST chaos/session-recovery harness, gateway REST surface (/v1/sessions, /v1/pool/utilization) + OpenAPI 3.0 — integration-proven vs real NATS/Kafka/HTTP
479efb5	Foundation wave 7: mark 9 items Completed (HXC-1101/1102/1103/1105/1106/1107/1114/1115/1133) — integration-proven vs real tmux/etcd/postgres
e12003a	Foundation wave 7: session WS-vertical (htmux attach+PTY-over-WS+msgpack envelope), e2ee ML-KEM record + sw-attestation, RBAC scopes, ratelimit token-bucket + etcd key namespace, postgres 15-table primary schema — integration-proven (real tmux/etcd/postgres); gate fixed tmux-CC subscribe race + pg applier host-path-in-container bug
6882b02	Foundation wave 6: mark 5 items Completed (HXC-1084/1085/1090/1095/1099)
1367dca	Foundation wave 6: session(tmux -CC + CRDT) / wireguard STUN-NAT / postgres registry / build orchestrator — integration-proven
e0d3938	Wave 5 deferred resolved: mark HXC-1086/1088/1093 Completed (integration-proven)

§2: Environment Snapshot

Property	Value
Branch	main
Commit	467a6dc
Timestamp	2026-06-01T17:28:49Z

§3: Active Work

HXC	Title	Status
HXC-922	Node service, advisory locks, chaos tests, docs	✅ Done
HXC-921	Web UI, K8s/Helm, integration, E2E, benchmarks	✅ Done
HXC-920		✅ Done

HXC	Title	Status
	Health, security, build, wireguard services	
HXC-919	htmux, messaging, GPU, WASM, storage, build manifest	✓ Done
HXC-918	Wire scheduler/session gRPC to real backends, etcd discovery	✓ Done
HXC-916	Stub expansion (config, retry, ratelimit, validator, build cache)	✓ Done
HXC-914	LLM, policy, setup, distributed lock	✓ Done
HXC-913	Health, security, ClassAds parser	✓ Done
HXC-912	Gateway, helixd, helix-agent	✓ Done
HXC-911	Scheduler, Session gRPC services	✓ Done
HXC-910	Console, security, testing infra	✓ Done

§4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation
4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

§5: Known Issues / Blockers

- None

§6: Quick Commands

```
# Run all tests (requires GOWORK due to workspace modules)
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

```
# Build all binaries
GOWORK=$(pwd)/go.work go build ./cmd/...
```

Build web UI

`cd web && npm run build`

Render Helm chart

`helm template helix-cluster deploy/helm/`

Run benchmarks

`go test -bench=. ./test/benchmark`

Run chaos tests

`go test -race -count=1 ./test/chaos`